FH fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes FH gene fusions from the msk_impact_50k_2026 study, covering 16 fusion events. 1/6 fusions with known frame status (16.7%) were in-frame. 2/15 fusions (13.3%) retained the Lyase domain. Domain retention was tested with Fisher's exact test (p=0.142857, not statistically significant at alpha=0.05). No literature benchmark is configured for FH.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 16 analyzed fusion events, identifying 11 distinct fusion partner genes. The most frequent partner was DGKH, observed in 5 events.

Domain retention

Lyase domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 1/1 (100.0%) of in-frame fusions retained the domain, p=0.142857 (not statistically significant at alpha=0.05). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.108911.

Domain disruption

Domain-disruption analysis was skipped: FH has no disruption_required_domains configured; domain-disruption analysis was skipped.

Junction concentration

67% of 6 fusions with this gene as the 5' partner join at exon 7, against 8% expected if breakpoints fell at random along the gene; this is the most over-represented junction (empirical p=0.00799201, corrected for testing every exon, statistically significant at alpha=0.05). The random-breakage null assumes a breakpoint anywhere in the gene would be detected. Capture panels such as MSK-IMPACT tile only selected introns, so concentration on panel data is partly by assay design and this p-value overstates the evidence for biological concentration. Confirm in a cohort sequenced without intron selection (whole genome or RNA).

Cutpoint detection

Cutpoint detection scanned 14 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 45 aa (exon 2; label-permutation p=0.0709291). This boundary is descriptive: retained/lost status follows from breakpoint position, so a clean split exists by construction and the p-value is not evidence of selection. This is -13 aa from the nearest configured domain boundary at 58 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" (n=2) and "not_retained" (n=12). For Frame_status, retained: 1/1 (100.0%, 95% CI 20.7%-100.0%); not_retained: 0/5

(0.0%, 95% CI 0.0%-43.4%). Welch's t-test comparing Tumor_variant_count between groups gave means of 12 vs 29.5, p=0.0809766 (not statistically significant at alpha=0.05).

Composite evidence score

Composite evidence score reported: domain_disruption sub-score excluded: the supplied result had no applicable p-value (e.g. disruption_required_domains not configured for this gene).

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Expression association

Expression association reported: No mRNA expression data was available for FH in this cohort; expression-association analysis was skipped.

Genomic position recurrence

Genomic position recurrence produced results for this run (8 row(s) in genomic_bin_recurrence, 0 row(s) in exact_position_recurrence, 2 row(s) in protein_position_genomic_spread); see the accompanying table.

Joint partner

Joint partner reported: FH has no gene_pair configured; joint-partner analysis was skipped.

Mechanistic interpretation

Mechanistic interpretation produced results for this run; see the accompanying data for details.

Mutation cooccurrence

Mutation cooccurrence reported: FH has no mutual_exclusivity_targets configured; co-occurrence/mutual-exclusivity analysis was skipped.

Window detection

Window detection produced results for this run (20 row(s) in window_scan, 11 row(s) in top_windows); see the accompanying table.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
DGKH	5	0.3125	0.09629286886244175	None	None	0.3860334247970314	confidence_certainty, domain_retention, recurrence	0.2406562456850978	1
RGS7	2	0.125	0.09629286886244175	None	None	0.3860334247970314	confidence_certainty, domain_retention, recurrence	0.15411778414663627	2
ARID4B	1	0.0625	0.09629286886244175	None	None	0.3860334247970314	confidence_certainty, domain_retention, recurrence	0.12527163030048244	3
FMN2	1	0.0625	0.09629286886244175	None	None	0.3860334247970314	confidence_certainty, domain_retention, recurrence	0.12527163030048244	4
MAFTRR	1	0.0625	0.09629286886244175	None	None	0.3860334247970314	confidence_certainty, domain_retention, recurrence	0.12527163030048244	5
NOTCH1	1	0.0625	0.09629286886244175	None	None	0.3860334247970314	confidence_certainty, domain_retention, recurrence	0.12527163030048244	6
PDE1C	1	0.0625	0.09629286886244175	None	None	0.3860334247970314	confidence_certainty, domain_retention, recurrence	0.12527163030048244	7
RYR2	1	0.0625	0.09629286886244175	None	None	0.3860334247970314	confidence_certainty, domain_retention, recurrence	0.12527163030048244	8
SLC35F3	1	0.0625	0.09629286886244175	None	None	0.3860334247970314	confidence_certainty, domain_retention, recurrence	0.12527163030048244	9
TRMT1L	1	0.0625	0.09629286886244175	None	None	0.3860334247970314	confidence_certainty, domain_retention, recurrence	0.12527163030048244	10
unknown	1	0.0625	0.09629286886244175	None	None	0.3860334247970314	confidence_certainty, domain_retention, recurrence	0.12527163030048244	11

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
44	0	2	1	11	0.0	1.0	-0.0

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
45	2	0	1	11	None	0.032967032967032975	1.481920137601431
186	2	0	2	10	None	0.06593406593406594	1.18089014193745
246	2	0	3	9	None	0.10989010989010989	0.9590413923210936
345	2	0	4	8	None	0.1648351648351648	0.7829501332654124
370	2	0	9	3	None	1.0	-0.0
464	2	0	10	2	None	1.0	-0.0
467	2	0	11	1	None	1.0	-0.0

domain_retention tables

frame_domain_contingency_table

1	1
0	12

frame_domain_contingency_tables_by_tumor_type

stratum	table
ANSC	[0, 0], [0, 1]
BRCA	[0, 1], [0, 0]
COAD	[0, 0], [0, 1]
CUP	[0, 0], [0, 1]
GBC	[0, 0], [0, 1]
HGSOC	[0, 0], [0, 1]
IDC	[0, 0], [0, 1]
LUAD	[0, 0], [0, 1]
LUAS	[1, 0], [0, 0]
LUSC	[0, 0], [0, 1]
MDLC	[0, 0], [0, 1]
PLSMESO	[0, 0], [0, 1]
READ	[0, 0], [0, 1]
SPDAC	[0, 0], [0, 1]

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
auto_key_domain	Lyase	14	2	8	4	0.4166666666666667

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
2	7	15	0.4666666666666667

frequency tables

Partner_gene_counts

Partner_gene	Event_count	Raw_event_count	Sample_count	Patient_count
ARID4B	1	1	1	1
DGKH	5	5	5	4
FMN2	1	1	1	1
MAFTRR	1	1	1	1
NOTCH1	1	1	1	1
PDE1C	1	1	1	1
RGS7	2	2	2	2
RYR2	1	1	1	1
SLC35F3	1	1	1	1
TRMT1L	1	1	1	1
unknown	1	1	1	1

genomic_position_recurrence tables

genomic_bin_recurrence

chromosome	bin_start	bin_end	n_events	event_ids
1	241667000	241668000	4	EVT-P-0018108-T01-IM6-8, EVT-P-0059293-T01-IM7-11, EVT-P-0064901-T01-IM7-9, EVT-P-0066922-T02-IM7-20
1	241661000	241662000	3	EVT-P-0051358-T01-IM6-29, EVT-P-0059435-T01-IM7-10, EVT-P-0060877-T01-IM7-21
1	241680000	241681000	2	EVT-P-0015961-T03-IM6-28, EVT-P-0048392-T01-IM6-1
1	241665000	241666000	1	EVT-P-0062340-T01-IM7-5

chromosome	bin_start	bin_end	n_events	event_ids
1	241666000	241667000	1	EVT-P-0004918-T02-IM6-26
1	241671000	241672000	1	EVT-P-0032745-T01-IM6-27
1	241672000	241673000	1	EVT-P-0040956-T01-IM6-22
1	241682000	241683000	1	EVT-P-0000532-T02-IM5-25

protein_position_genomic_spread

protein_position_aa	n_events	n_distinct_genomic_positions	genomic_span_bp	spans_multiple_chromosomes
370	5	5	1445	False
45	2	2	101	False

junction_concentration tables

junction_counts

role	junction_exon	count	role_event_count	observed_share	random_breakage_share	excess_neg_log10_p
five_prime	7	4	6	0.6666666666666666	0.07559005370278442	3.3641557507141013
three_prime	10	3	8	0.375	0.12789385802608422	1.1469025228149392
three_prime	2	2	8	0.25	0.10871429216119861	0.6712630418201881
three_prime	8	2	8	0.25	0.07216029604224017	0.9626118450956878
five_prime	1	1	6	0.16666666666666666	0.1103389142109301	0.2974375245674
five_prime	5	1	6	0.16666666666666666	0.11810099733742498	0.2760927041584447
three_prime	5	1	8	0.125	0.1518119048693533	0.13541666429171684

window_detection tables

window_scan

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
-14	186	200	4	10	2	0	None	0.06593406593406594	1.18089014193745	EVT-P-0000532-T02-IM5-25, EVT-P-0015961-T03-IM6-28, EVT-P-0040956-T01-IM6-22, EVT-P-0048392-T01-IM6-1
44	244	200	4	10	2	0	None	0.06593406593406594	1.18089014193745	EVT-P-0000532-T02-IM5-25, EVT-P-0015961-T03-IM6-28, EVT-P-0040956-T01-IM6-22, EVT-P-0048392-T01-IM6-1

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
170	370	200	8	6	0	2	0.0	0.16483516483516483	0.7829501332654124	EVT-P-0004918-T02-IM6-26, EVT-P-0018108-T01-IM6-8, EVT-P-0032745-T01-IM6-27, EVT-P-0040956-T01-IM6-22, EVT-P-0059293-T01-IM7-11, ... (8 total)
186	386	200	8	6	0	2	0.0	0.16483516483516483	0.7829501332654124	EVT-P-0004918-T02-IM6-26, EVT-P-0018108-T01-IM6-8, EVT-P-0032745-T01-IM6-27, EVT-P-0040956-T01-IM6-22, EVT-P-0059293-T01-IM7-11, ... (8 total)
246	446	200	7	7	0	2	0.0	0.46153846153846156	0.3357921019231931	EVT-P-0004918-T02-IM6-26, EVT-P-0018108-T01-IM6-8, EVT-P-0032745-T01-IM6-27, EVT-P-0059293-T01-IM7-11, EVT-P-0062340-T01-IM7-5, ... (7 total)
264	464	200	7	7	0	2	0.0	0.46153846153846156	0.3357921019231931	EVT-P-0004918-T02-IM6-26, EVT-P-0018108-T01-IM6-8, EVT-P-0059293-T01-IM7-11, EVT-P-0059435-T01-IM7-10, EVT-P-0062340-T01-IM7-5, ... (7 total)
267	467	200	8	6	0	2	0.0	0.16483516483516483	0.7829501332654124	EVT-P-0004918-T02-IM6-26, EVT-P-0018108-T01-IM6-8, EVT-P-0059293-T01-IM7-11, EVT-P-0059435-T01-IM7-10, EVT-P-0060877-T01-IM7-21, ... (8 total)
270	370	100	6	8	0	2	0.0	0.4725274725274724	0.32557293674150717	EVT-P-0004918-T02-IM6-26, EVT-P-0018108-T01-IM6-8, EVT-P-0059293-T01-IM7-11, EVT-P-0062340-T01-IM7-5, EVT-P-0064901-T01-IM7-9, ... (6 total)
277	477	200	9	5	0	2	0.0	0.10989010989010989	0.9590413923210936	EVT-P-0004918-T02-IM6-26, EVT-P-0018108-T01-IM6-8, EVT-P-0051358-T01-IM6-29, EVT-P-0059293-T01-IM7-11, EVT-P-0059435-T01-IM7-10, ... (9 total)
320	370	50	6	8	0	2	0.0	0.4725274725274724	0.32557293674150717	EVT-P-0004918-T02-IM6-26, EVT-P-0018108-T01-IM6-8, EVT-P-0059293-T01-IM7-11, EVT-P-0062340-T01-IM7-5, EVT-P-0064901-T01-IM7-9, ... (6 total)
345	370	25	6	8	0	2	0.0	0.4725274725274724	0.32557293674150717	EVT-P-0004918-T02-IM6-26, EVT-P-0018108-T01-IM6-8, EVT-P-0059293-T01-IM7-11, EVT-P-0062340-T01-IM7-5, EVT-P-0064901-T01-IM7-9, ... (6 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
345	395	50	6	8	0	2	0.0	0.4725274725274724	0.32557293674150717	EVT-P-0004918-T02-IM6-26, EVT-P-0018108-T01-IM6-8, EVT-P-0059293-T01-IM7-11, EVT-P-0062340-T01-IM7-5, EVT-P-0064901-T01-IM7-9, ... (6 total)
345	445	100	6	8	0	2	0.0	0.4725274725274724	0.32557293674150717	EVT-P-0004918-T02-IM6-26, EVT-P-0018108-T01-IM6-8, EVT-P-0059293-T01-IM7-11, EVT-P-0062340-T01-IM7-5, EVT-P-0064901-T01-IM7-9, ... (6 total)
345	545	200	9	5	0	2	0.0	0.10989010989010989	0.9590413923210936	EVT-P-0004918-T02-IM6-26, EVT-P-0018108-T01-IM6-8, EVT-P-0051358-T01-IM6-29, EVT-P-0059293-T01-IM7-11, EVT-P-0059435-T01-IM7-10, ... (9 total)
364	464	100	6	8	0	2	0.0	0.4725274725274724	0.32557293674150717	EVT-P-0004918-T02-IM6-26, EVT-P-0018108-T01-IM6-8, EVT-P-0059293-T01-IM7-11, EVT-P-0059435-T01-IM7-10, EVT-P-0062340-T01-IM7-5, ... (6 total)
367	467	100	7	7	0	2	0.0	0.46153846153846156	0.3357921019231931	EVT-P-0004918-T02-IM6-26, EVT-P-0018108-T01-IM6-8, EVT-P-0059293-T01-IM7-11, EVT-P-0059435-T01-IM7-10, EVT-P-0060877-T01-IM7-21, ... (7 total)
370	395	25	5	9	0	2	0.0	0.5054945054945055	0.29628356063951955	EVT-P-0004918-T02-IM6-26, EVT-P-0018108-T01-IM6-8, EVT-P-0059293-T01-IM7-11, EVT-P-0062340-T01-IM7-5, EVT-P-0064901-T01-IM7-9
370	420	50	5	9	0	2	0.0	0.5054945054945055	0.29628356063951955	EVT-P-0004918-T02-IM6-26, EVT-P-0018108-T01-IM6-8, EVT-P-0059293-T01-IM7-11, EVT-P-0062340-T01-IM7-5, EVT-P-0064901-T01-IM7-9
370	470	100	7	7	0	2	0.0	0.46153846153846156	0.3357921019231931	EVT-P-0004918-T02-IM6-26, EVT-P-0018108-T01-IM6-8, EVT-P-0059293-T01-IM7-11, EVT-P-0059435-T01-IM7-10, EVT-P-0060877-T01-IM7-21, ... (7 total)
370	570	200	8	6	0	2	0.0	0.16483516483516483	0.7829501332654124	EVT-P-0004918-T02-IM6-26, EVT-P-0018108-T01-IM6-8, EVT-P-0051358-T01-IM6-29, EVT-P-0059293-T01-IM7-11, EVT-P-0059435-T01-IM7-10, ... (8 total)

top_windows

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
-14	186	200	4	10	2	0	None	0.06593406593406594	1.18089014193745	EVT-P-0000532-T02-IM5-25, EVT-P-0015961-T03-IM6-28, EVT-P-0040956-T01-IM6-22, EVT-P-0048392-T01-IM6-1
277	477	200	9	5	0	2	0.0	0.10989010989010989	0.9590413923210936	EVT-P-0004918-T02-IM6-26, EVT-P-0018108-T01-IM6-8, EVT-P-0051358-T01-IM6-29, EVT-P-0059293-T01-IM7-11, EVT-P-0059435-T01-IM7-10, ... (9 total)
170	370	200	8	6	0	2	0.0	0.16483516483516483	0.7829501332654124	EVT-P-0004918-T02-IM6-26, EVT-P-0018108-T01-IM6-8, EVT-P-0032745-T01-IM6-27, EVT-P-0040956-T01-IM6-22, EVT-P-0059293-T01-IM7-11, ... (8 total)
267	467	200	8	6	0	2	0.0	0.16483516483516483	0.7829501332654124	EVT-P-0004918-T02-IM6-26, EVT-P-0018108-T01-IM6-8, EVT-P-0059293-T01-IM7-11, EVT-P-0059435-T01-IM7-10, EVT-P-0060877-T01-IM7-21, ... (8 total)
370	570	200	8	6	0	2	0.0	0.16483516483516483	0.7829501332654124	EVT-P-0004918-T02-IM6-26, EVT-P-0018108-T01-IM6-8, EVT-P-0051358-T01-IM6-29, EVT-P-0059293-T01-IM7-11, EVT-P-0059435-T01-IM7-10, ... (8 total)
367	467	100	7	7	0	2	0.0	0.46153846153846156	0.3357921019231931	EVT-P-0004918-T02-IM6-26, EVT-P-0018108-T01-IM6-8, EVT-P-0059293-T01-IM7-11, EVT-P-0059435-T01-IM7-10, EVT-P-0060877-T01-IM7-21, ... (7 total)
246	446	200	7	7	0	2	0.0	0.46153846153846156	0.3357921019231931	EVT-P-0004918-T02-IM6-26, EVT-P-0018108-T01-IM6-8, EVT-P-0032745-T01-IM6-27, EVT-P-0059293-T01-IM7-11, EVT-P-0062340-T01-IM7-5, ... (7 total)
264	464	200	7	7	0	2	0.0	0.46153846153846156	0.3357921019231931	EVT-P-0004918-T02-IM6-26, EVT-P-0018108-T01-IM6-8, EVT-P-0059293-T01-IM7-11, EVT-P-0059435-T01-IM7-10, EVT-P-0062340-T01-IM7-5, ... (7 total)
345	370	25	6	8	0	2	0.0	0.4725274725274724	0.32557293674150717	EVT-P-0004918-T02-IM6-26, EVT-P-0018108-T01-IM6-8, EVT-P-0059293-T01-IM7-11, EVT-P-0062340-T01-IM7-5, EVT-P-0064901-T01-IM7-9, ... (6 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
364	464	100	6	8	0	2	0.0	0.4725274725274724	0.32557293674150717	EVT-P-0004918-T02-IM6-26, EVT-P-0018108-T01-IM6-8, EVT-P-0059293-T01-IM7-11, EVT-P-0059435-T01-IM7-10, EVT-P-0062340-T01-IM7-5, ... (6 total)
370	395	25	5	9	0	2	0.0	0.5054945054945055	0.29628356063951955	EVT-P-0004918-T02-IM6-26, EVT-P-0018108-T01-IM6-8, EVT-P-0059293-T01-IM7-11, EVT-P-0062340-T01-IM7-5, EVT-P-0064901-T01-IM7-9

Per-event results (results.tsv)

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-004839 2-T01-IM6-1	P-004839 2	P-004839 2	Breast Cancer	BRCA		unknown	unknown	three_prime	241680684	2	45	True	45	True	exact match	retained	1.0	False												Lyase			FH (NM_000143) rearrangement: c.133-68:FH_chr1:g.240716629inv | Transcript Fusion {MIR1273E:FH}	NA
EVT-P-006234 0-T01-IM7-5	P-006234 0	P-006234 0	Non-Small Cell Lung Cancer	LUAD	ARID4B-FH	ARID4B	out-of-frame	three_prime	241665896	8	370	True	370	True	exact match	disrupted	0.060240963855421686	True	partner_lacks_domain	PF00206					614			False				Lyase	ARID4B (NM_016374) - FH (NM_000143) rearrangement: c.1841+5245:ARID4B_c.1109-26:FHdup | Protein Fusion: out of frame {ARID4B:FH}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001810 8-T02-IM6-7	P-001810 8-T02-IM6	P-001810 8	Non-Small Cell Lung Cancer	LUSC	DGKH-FH	DGKH	unknown	five_prime	241667340	7	370	True	369	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.942771084337494	True	partner_lacks_domain	PF00206						1221			False				Lyase	FH (NM_000143) rearrangement: t(1;13)(q43;q14.11)(chr1:g.241667340::chr13:g.42830153)|Transcript Fusion {FH:DGKH}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001810 8-T01-IM6-8	P-001810 8-T01-IM6	P-001810 8	Non-Small Cell Lung Cancer	LUSC	DGKH-FH	DGKH	unknown	five_prime	241667339	7	370	True	369	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.942771084337494	True	partner_lacks_domain	PF00206					1221			False				Lyase	FH (NM_000143) rearrangement: t(1;13)(q43;q14.11)(chr1:g.241667340::chr13:g.42830153)|Transcript Fusion {FH:DGKH}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-006490 1-T01-IM7-9	P-006490 1-T01-IM7	P-006490 1	Ovarian Cancer	HGSOC	DGKH-FH	DGKH	unknown	five_prime	241667338	7	370	True	369	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.942771084337349 4	True	partner_lacks_domain	PF00206					1221			False				Lyase	FH (NM_000143) rearrangement: t(1;13)(q43;q14.11)(chr1:g.241667338::chr13:g.42830159)|Transcript Fusion {FH:DGKH}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-005943 5-T01-IM7-10	P-005943 5-T01-IM7	P-005943 5	Esophagogastric Cancer	SPDAC	DGKH-FH	DGKH	unknown	three_prime	241661272	10	464	True	464	True	exact match	lost	0.0	False													Lyase		FH (NM_000143) rearrangement: t(1;13)(q43;q14.11)(chr1:g.241661272::chr13:g.42830525)|Transcript Fusion {DGKH:FH}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-005923-T01-IM7-11	P-005923-T01-IM7	P-005923	Cancer of Unknown Primary	CUP	DGKH-FH	DGKH	unknown	five_prime	241667341	7	370	True	369	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.9427710843373494	True	partner_lacks_domain	PF00206					1221			False				Lyase	FH (NM_000143) rearrangement: t(1;13)(q43;q14.11)(chr1:g.241667341::chr13:g.42830159)|Protein Fusion: mid-exon {FH:DGKH}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-006692-T02-IM7-20	P-006692-T02-IM7	P-006692	Hepatobiliary Cancer	GBC	FMN2-FH	FMN2	unknown	five_prime	241667416	7	345	False	344	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.8674698795180723	True	partner_lacks_domain	PF00206					1406			False				Lyase	FH (NM_000143) rearrangement: c.1034:FH_chr1:g.240469632del | Antisense Fusion	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-006087-T01-IM7-21	P-006087-T01-IM7	P-006087	Colorectal Cancer	COAD	MAFTRR-FH	MAFTRR	unknown	three_prime	241661260	10	467	False	467	True	exact match	lost	0.0	False													Lyase		FH (NM_000143) rearrangement: t(1;16)(q43;q23.2)(chr1:g.241661260::chr16:g.79754421)|Protein Fusion: mid-exon {LOC102467146:FH}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-004096-T01-IM6-22	P-004096-T01-IM6	P-004096	Anal Cancer	ANSC	NOTCH1-FH	NOTCH1	out-of-frame	three_prime	241672198	5	186	True	186	True	exact match	disrupted	0.6144578313253012	True	partner_lacks_domain	PF00206					135			False				Lyase	NOTCH1 (NM_017617) - FH (NM_000143) rearrangement: t(1;9)(q43;q34.3)(chr1:g.241672198::chr9:g.139417940) | Protein Fusion: out of frame {NOTCH1: FH}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-000532-T02-IM5-25	P-000532-T02-IM5	P-000532	Breast Cancer	MDLC	PDE1C-FH	PDE1C	out-of-frame	five_prime	241682866	1	44	True	44	True	exact match	lost	0.0	False													Lyase		FH (NM_000143) - PDE1C (NM_001191058) Rearrangement : t(1;7)(1q43;7p14.3)(chr1:g.241682866::chr7:g.31926696) | Protein fusion: out of frame (FH-PDE1C)	NA

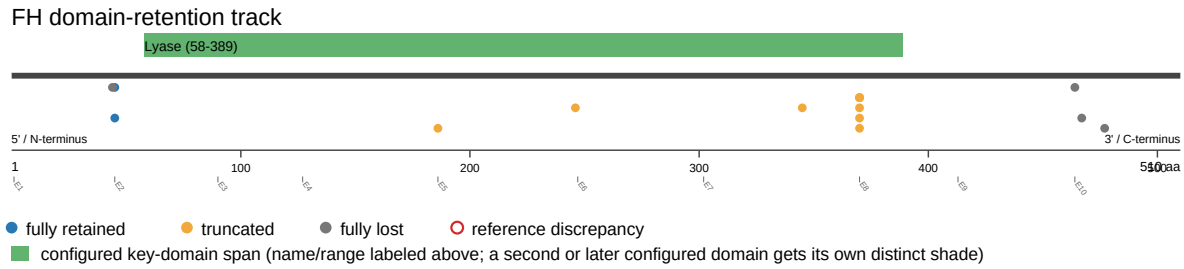
event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-000491 8-T02-IM6-26	P-000491 8-T02-IM6	P-000491 8	Breast Cancer	IDC	RGS7-FH	RGS7	out-of-frame	three_prime	241666009	8	370	True	370	True	exact match	disrupted	0.060240963855421686	True	partner_lacks_domain	PF00206					26			False				Lyase	RGS7 (NM_002924) - FH (NM_000143) rearrangement: c.79-95591:RGS7_c.1109-139:FHdup | Protein Fusion: out of frame {RGS7:FH}	NA

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-003274 5-T01-IM6-27	P-003274 5-T01-IM6	P-003274 5	Mesothelioma	PLSMESO	RGS7-FH	RGS7	out-of-frame	five_prime	241671722	5	246	True	246	True	exact match	disrupted	0.569277108433734 9	True	partner_lacks_domain	PF00206					129			False				Lyase	FH (NM_000143) - RGS7 (NM_002924) rearrangement: c.738+181:FH_c.386-8492:RGS7del | Protein Fusion: out of frame {FH:RGS7}	NA
EVT-P-001596 1-T03-IM6-28	P-001596 1-T03-IM6	P-001596 1	Non-Small Cell Lung Cancer	LUAS	RYR2-FH	RYR2	in-frame	three_prime	241680785	2	45	True	45	True	exact match	retained	1.0	False												Lyase			RYR2 (NM_001035) - FH (NM_000143) rearrangement: c.48+56557:RYR2_c.133-169:FHinv | ProteinFusion: in frame {RYR2:FH}	NA

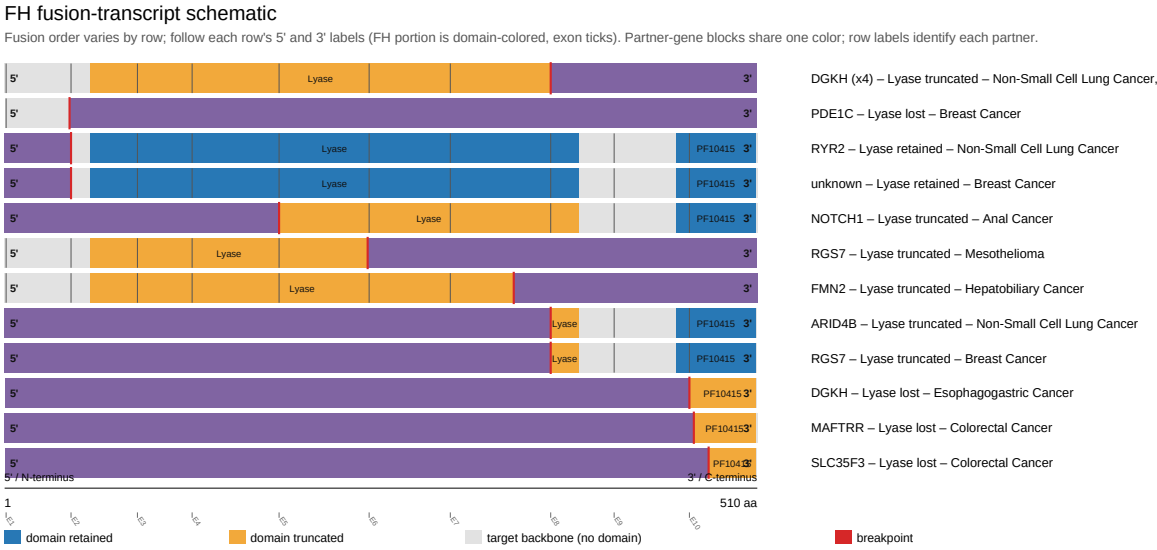
event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-005135 8-T01-IM6-29	P-005135 8-T01-IM6	P-005135 8	Colorectal Cancer	READ	SLC35F3-FH	SLC35F3	unknown	three_prime	241661231	10	477	False	477	True	exact match	lost	0.0	False															Lyase		SLC35F3 (NM_173508) - FH (NM_000143) rearrangement: c.283+157829:SLC35F3_c.1430:FHinv | Protein Fusion: mid-exon {SLC35F3:FH}	NA

Figures

domain retention outliers



fusion schematic



Partner-block colors are decorative; row labels identify partners.